LEARNING MODULE OVERVIEW

- The expected value of a random variable is a probability-weighted average of the possible outcomes of the random variable. For a random variable X, the expected value of X is denoted $E(X)$.
- The variance of a random variable is the expected value (the probability-weighted average) of squared deviations from the random variable's expected value $E(X)$: $\sigma^2(X) = E\{[X - E(X)]^2\}$, where $\sigma^2(X)$ stands for the variance of X.
- Standard deviation is the positive square root of variance. Standard deviation measures dispersion (as does variance), but it is measured in the same units as the variable.
- A probability tree is a means of illustrating the results of two or more independent events.
- A probability of an event given (conditioned on) another event is a conditional probability. The probability of an event A given an event B is denoted $P(A \mid B)$, and $P(A \mid B) = P(AB)/P(B)$, $P(B) \neq 0$.
- According to the total probability rule, if $S1$, $S2$, …, Sn are mutually exclusive and exhaustive scenarios or events, then $P(A) = P(A \mid S1)P(S1) + P(A \mid S2)P(S2) + \ldots + P(A \mid Sn)P(Sn)$.
- Conditional expected value is $E(X \mid S) = P(X_1 \mid S)X_1 + P(X_2 \mid S)X_2 + \ldots + P(X_n \mid S)X_n$ and has an associated conditional variance and conditional standard deviation.
- Bayes' formula is a method used to update probabilities based on new information.
- Bayes' formula is expressed as follows: Updated probability of event given the new information = [(Probability of the new information given event)/(Unconditional probability of the new information)] × Prior probability of event.

2 EXPECTED VALUE AND VARIANCE

☐ calculate expected values, variances, and standard deviations and demonstrate their application to investment problems

The expected value of a random variable is an essential quantitative concept in investments. Investors continually make use of expected values—in estimating the rewards of alternative investments, in forecasting earnings per share (EPS) and other corporate financial variables and ratios, and in assessing any other factor that may affect their financial position. The **expected value of a random variable** is the probability-weighted average of the possible outcomes of the random variable. For a random variable X, the expected value of X is denoted $E(X)$.

Expected value (e.g., expected stock return) looks either to the future, as a forecast, or to the "true" value of the mean (the population mean). We should distinguish expected value from the concepts of historical or sample mean. The sample mean also